BSDS Exercise Set 2

Detailed step-by-step LaTeX solutions

Note on the OCR text in Question 1(b). The uploaded PDF repeats the symbol $P(\xi = 2)$ twice. In the solution below, I interpret this as the intended offspring distribution

$$\mathbb{P}(\xi = 0) = \frac{1}{6}, \qquad \mathbb{P}(\xi = 1) = \frac{1}{2}, \qquad \mathbb{P}(\xi = 2) = \frac{1}{6}, \qquad \mathbb{P}(\xi = 3) = \frac{1}{6},$$

so that the probabilities sum to 1. The method is the same for any corrected version of the typo.

Question

Problem 1. Consider a branching process starting with a single individual, $Z_0 = 1$.

(a) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = \frac{1}{4}$, $\mathbb{P}(\xi = 1) = \frac{1}{2}$, $\mathbb{P}(\xi = 2) = \frac{1}{4}$, compute $\mathbb{E}(Z_3)$ and $\mathrm{Var}(Z_3)$. Also, find the probability that $\mathbb{P}(Z_3 = 0)$.

(b) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = \frac{1}{6}$, $\mathbb{P}(\xi = 1) = \frac{1}{2}$, $\mathbb{P}(\xi = 2) = \frac{1}{6}$, $\mathbb{P}(\xi = 3) = \frac{1}{6}$, compute $\mathbb{E}(Z_2 Z_3)$ and $\mathrm{Var}(Z_2 + Z_3)$. Also, find the probability $\mathbb{P}(Z_2 > 0, Z_3 = 0)$.

Theory used

For a Galton–Watson branching process with offspring pgf $G(s) = \mathbb{E}(s^\xi)$, the pgf of the n-th generation is the n-fold iterate $G_n = G \circ G \circ \cdots \circ G$. Also,

$$\mathbb{E}(Z_n) = \mu^n, \qquad \mu = \mathbb{E}(\xi),$$

and if $\sigma^2 = \mathrm{Var}(\xi) < \infty$, then

$$\mathrm{Var}(Z_n) = \begin{cases} n\sigma^2, & \mu = 1, \\ \sigma^2 \mu^{n-1} \dfrac{\mu^n - 1}{\mu - 1}, & \mu \neq 1. \end{cases}$$

Moreover, $\mathbb{P}(Z_n = 0) = G_n(0)$.

Step-by-step solution

(a) Step 1: Write the offspring pgf.
The offspring distribution is

$$G(s) = \frac{1}{4} + \frac{1}{2}s + \frac{1}{4}s^2.$$

It is convenient to factor it:

$$G(s) = \frac{(1+s)^2}{4}.$$

Step 2: Compute the mean and variance of the offspring distribution.
First,

$$\mu = \mathbb{E}(\xi) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1.$$

Next,

$$\mathbb{E}(\xi^2) = 0^2 \cdot \frac{1}{4} + 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{4} = \frac{1}{2} + 1 = \frac{3}{2}.$$

Hence

$$\sigma^2 = \mathrm{Var}(\xi) = \mathbb{E}(\xi^2) - \mu^2 = \frac{3}{2} - 1 = \frac{1}{2}.$$

Step 3: Compute $\mathbb{E}(Z_3)$.
For a branching process,

$$\mathbb{E}(Z_n) = \mu^n.$$

Since $\mu = 1$,

$$\mathbb{E}(Z_3) = 1^3 = 1.$$

Step 4: Compute $\mathrm{Var}(Z_3)$.
Because $\mu = 1$, the variance formula becomes

$$\mathrm{Var}(Z_n) = n\sigma^2.$$

Therefore,

$$\mathrm{Var}(Z_3) = 3 \cdot \frac{1}{2} = \frac{3}{2}.$$

Step 5: Compute $\mathbb{P}(Z_3 = 0)$.
The extinction probability at time 3 is

$$\mathbb{P}(Z_3 = 0) = G_3(0) = G(G(G(0))).$$

Now compute this step by step:

$$G(0) = \frac{1}{4},$$

$$G_2(0) = G\Big(\frac{1}{4}\Big) = \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{16} = \frac{1}{4} + \frac{1}{8} + \frac{1}{64} = \frac{25}{64},$$

and then

$$G_3(0) = G\Big(\frac{25}{64}\Big) = \frac{1}{4} + \frac{1}{2} \cdot \frac{25}{64} + \frac{1}{4} \cdot \Big(\frac{25}{64}\Big)^2.$$

Putting everything over a common denominator,

$$G_3(0) = \frac{1}{4} + \frac{25}{128} + \frac{625}{16384} = \frac{4096 + 3200 + 625}{16384} = \frac{7921}{16384}.$$

Theory used

For branching processes, conditioning on the present generation gives

$$\mathbb{E}(Z_{n+1} \mid Z_n) = \mu Z_n.$$

This implies

$$\mathbb{E}(Z_n Z_m) = \mu^{n-m}\mathbb{E}(Z_m^2), \qquad m \le n,$$

and therefore

$$\mathrm{Cov}(Z_m, Z_n) = \mu^{n-m}\mathrm{Var}(Z_m).$$

Also, if $n > m$, then

$$\mathbb{E}(Z_n \mid Z_m = k) = k\mu^{n-m}.$$

Step-by-step solution

(b) Step 1: Identify the offspring pgf.
With the corrected interpretation,

$$G(s) = \frac{1}{6} + \frac{1}{2}s + \frac{1}{6}s^2 + \frac{1}{6}s^3.$$

Step 2: Compute the mean and variance of one offspring.
We calculate

$$\mu = \mathbb{E}(\xi) = 0 \cdot \frac{1}{6} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} = \frac{1}{2} + \frac{1}{3} + \frac{1}{2} = \frac{4}{3}.$$

Also,

$$\mathbb{E}(\xi^2) = 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{6} + 3^2 \cdot \frac{1}{6} = \frac{1}{2} + \frac{4}{6} + \frac{9}{6} = \frac{8}{3}.$$

Hence

$$\sigma^2 = \mathrm{Var}(\xi) = \frac{8}{3} - \left(\frac{4}{3}\right)^2 = \frac{8}{3} - \frac{16}{9} = \frac{8}{9}.$$

Step 3: Compute $\mathbb{E}(Z_2 Z_3)$.
Use

$$\mathbb{E}(Z_3 \mid Z_2) = \mu Z_2.$$

So

$$\mathbb{E}(Z_2 Z_3) = \mathbb{E}\Big(Z_2\,\mathbb{E}(Z_3 \mid Z_2)\Big) = \mathbb{E}(Z_2 \cdot \mu Z_2) = \mu\,\mathbb{E}(Z_2^2).$$

Now

$$\mathbb{E}(Z_2) = \mu^2 = \left(\frac{4}{3}\right)^2 = \frac{16}{9}.$$

The variance at generation 2 is

$$\mathrm{Var}(Z_2) = \sigma^2 \mu \frac{\mu^2 - 1}{\mu - 1} = \frac{8}{9} \cdot \frac{4}{3} \cdot \frac{\frac{16}{9} - 1}{\frac{4}{3} - 1} = \frac{224}{81}.$$

Hence

$$\mathbb{E}(Z_2^2) = \mathrm{Var}(Z_2) + \mathbb{E}(Z_2)^2 = \frac{224}{81} + \left(\frac{16}{9}\right)^2 = \frac{224}{81} + \frac{256}{81} = \frac{160}{27}.$$

Therefore,

$$\mathbb{E}(Z_2 Z_3) = \frac{4}{3} \cdot \frac{160}{27} = \frac{640}{81}.$$

Step 4: Compute $\mathrm{Var}(Z_2 + Z_3)$.
We use

$$\mathrm{Var}(Z_2 + Z_3) = \mathrm{Var}(Z_2) + \mathrm{Var}(Z_3) + 2\mathrm{Cov}(Z_2, Z_3).$$

We already have $\mathrm{Var}(Z_2) = 224/81$. Also,

$$\mathrm{Var}(Z_3) = \sigma^2 \mu^2 \frac{\mu^3 - 1}{\mu - 1} = \frac{8}{9} \cdot \left(\frac{4}{3}\right)^2 \cdot \frac{\left(\frac{4}{3}\right)^3 - 1}{\frac{4}{3} - 1} = \frac{4736}{729}.$$

Next,

$$\mathrm{Cov}(Z_2, Z_3) = \mu^{3-2}\mathrm{Var}(Z_2) \overset{4}{=} \mu\,\mathrm{Var}(Z_2) = \frac{4}{3} \cdot \frac{224}{81} = \frac{896}{243}.$$

Thus

$$\mathrm{Var}(Z_2 + Z_3) = \frac{224}{} + \frac{4736}{} + 2 \cdot \frac{896}{} = \frac{12128}{}$$

Question

Problem 2. Suppose that $\{Z_n : n \geq 0\}$ is a branching process with $Z_0 = 1$. Fix any positive integer $r \geq 1$ and define $\{Y_n : n \geq 0\}$ by

$$Y_n = Z_{nr}, \qquad n \geq 0.$$

Check that $\{Y_n\}$ is also a branching process. Find its probability generating function of the offspring distribution.

Theory used

If G is the offspring pgf of a branching process, then the pgf of the n-th generation is the n-fold iterate $G_n = G^{\circ n}$. In particular, the evolution over r generations acts like one new generation whose offspring pgf is G_r.

Step-by-step solution

Step 1: Interpret Y_n as a coarse-grained version of Z_n.
The variable $Y_n = Z_{nr}$ records the population every r-th generation only. So one step of $\{Y_n\}$ corresponds to r steps of the original branching process.

Step 2: Use the branching property.
Starting from one individual, the number of individuals after r generations has pgf

$$G_r(s) = \underbrace{G(G(\cdots G(s)\cdots))}_{r \text{ times}}.$$

Thus, if one individual of the Y-process produces the population after r generations, then its offspring pgf is exactly G_r.

Step 3: Verify the branching structure.
If $Y_n = k$, then there are k independent families at time nr, and each family evolves independently for the next r generations. Hence the next state Y_{n+1} is the sum of k independent copies of the r-generation offspring count. That is precisely the branching property.

Conclusion.
Therefore $\{Y_n\}$ is also a branching process, and its offspring pgf is

$$\boxed{G_Y(s) = G_r(s) = G^{\circ r}(s).}$$

Question

Problem 3. Consider a branching process with offspring pgf

$$G(s) = \frac{q}{1-ps}, \qquad 0 < p < 1, \quad q = 1-p.$$

(a) For $p = \frac{1}{2}$, show that

$$G_n(s) = \frac{n-(n-1)s}{n+1-ns}, \qquad n \geq 1,$$

and show that $G_n(0) \to 1$ as $n \to \infty$.

(b) For $p \neq \frac{1}{2}$, derive an explicit formula for $G_n(s)$. Further conclude that

$$\lim_{n\to\infty} G_n(0) = \begin{cases} 1, & p < q, \\ \frac{q}{p}, & p > q. \end{cases}$$

Theory used

For a branching process, the pgf at generation n is the n-fold iterate $G_n = G^{\circ n}$. Also, the extinction probability is the smallest fixed point of $G(s) = s$ in $[0, 1]$. When G is linear fractional, iterates can be found by induction or by a change of variables that turns composition into multiplication.

Step-by-step solution

(a) Step 1: Specialize the pgf.
If $p = q = \frac{1}{2}$, then

$$G(s) = \frac{1/2}{1 - \frac{1}{2}s} = \frac{1}{2 - s}.$$

Step 2: Guess the form and verify by induction.
We claim

$$G_n(s) = \frac{n - (n-1)s}{n + 1 - ns}, \qquad n \geq 1.$$

For $n = 1$,

$$G_1(s) = G(s) = \frac{1}{2 - s},$$

and this is exactly

$$\frac{1 - (1-1)s}{1 + 1 - 1 \cdot s} = \frac{1}{2 - s}.$$

Assume the formula holds for n. Then

$$G_{n+1}(s) = G(G_n(s)) = \frac{1}{2 - G_n(s)}.$$

Substitute the inductive hypothesis:

$$G_{n+1}(s) = \frac{1}{2 - \frac{n-(n-1)s}{n+1-ns}}.$$

Put over a common denominator:

$$2 - \frac{n - (n-1)s}{n + 1 - ns} = \frac{2(n + 1 - ns) - (n - (n-1)s)}{n + 1 - ns} = \frac{n + 2 - (n+1)s}{n + 1 - ns}.$$

Therefore

$$G_{n+1}(s) = \frac{n + 1 - ns}{n + 2 - (n+1)s} = \frac{(n+1) - ns}{(n+1) + 1 - (n+1)s},$$

which is the same formula with n replaced by $n + 1$. Hence the claim is proved.

Step 3: Evaluate $G_n(0)$.
Setting $s = 0$,

$$G_n(0) = \frac{n}{n+1} \xrightarrow[n\to\infty]{} 1.$$

Conclusion for (a):

$$\boxed{G_n(s) = \frac{n - (n-1)s}{n + 1 - ns}, \qquad G_n(0) \to 1.}$$

(b) Step 1: Find a useful fixed-point transform.
Let

$$r = \frac{q}{p}.$$

Then r is the nontrivial fixed point of the map $G(s)$, because

$$s = \frac{q}{1 - ps} \iff ps^2 - s + q = 0 \iff (s-1)\left(s - \frac{q}{p}\right) = 0.$$

Question

Problem 4. Let $\{Z_n\}$ be a branching process with $Z_0 = 1$ and offspring mean μ.

(a) Assume that $G''(1) < \infty$. Show that

$$\mathbb{E}(Z_n Z_m) = \mu^{n-m}\mathbb{E}(Z_m^2), \qquad m \le n.$$

Hence find the covariance and correlation coefficient between Z_m and Z_n (in terms of μ) for $m \le n$.

(b) Find $\mathbb{E}(Z_n \mid Z_m = k)$ for $m < n$ and $k \ge 0$.

Theory used

For a branching process, the conditional mean is

$$\mathbb{E}(Z_n \mid Z_m) = \mu^{n-m} Z_m, \qquad n \ge m.$$

Also, if $G''(1) < \infty$, then $\mathrm{Var}(Z_n)$ is finite and can be written in terms of μ and $\sigma^2 = \mathrm{Var}(\xi)$. The law of total expectation and the law of total variance are the main tools.

Step-by-step solution

(a) Step 1: Compute the conditional mean of later generations.
Given Z_m, the process from generation m onward consists of Z_m independent families. Each family has expected size μ^{n-m} at generation n. Hence

$$\mathbb{E}(Z_n \mid Z_m) = \mu^{n-m} Z_m.$$

Step 2: Multiply by Z_m and take expectation.
We get

$$\mathbb{E}(Z_n Z_m) = \mathbb{E}\Big(Z_m \, \mathbb{E}(Z_n \mid Z_m)\Big) = \mathbb{E}\Big(Z_m \cdot \mu^{n-m} Z_m\Big) = \mu^{n-m}\mathbb{E}(Z_m^2).$$

This proves the identity.

Step 3: Compute the covariance.
Since $\mathbb{E}(Z_n) = \mu^n$ and $\mathbb{E}(Z_m) = \mu^m$,

$$\mathrm{Cov}(Z_m, Z_n) = \mathbb{E}(Z_m Z_n) - \mathbb{E}(Z_m)\mathbb{E}(Z_n) = \mu^{n-m}\mathbb{E}(Z_m^2) - \mu^{m+n}.$$

Now write $\mathbb{E}(Z_m^2) = \mathrm{Var}(Z_m) + \mu^{2m}$. Then

$$\mathrm{Cov}(Z_m, Z_n) = \mu^{n-m}\Big(\mathrm{Var}(Z_m) + \mu^{2m}\Big) - \mu^{m+n} = \mu^{n-m}\mathrm{Var}(Z_m).$$

Step 4: Correlation coefficient.
Therefore

$$\mathrm{Corr}(Z_m, Z_n) = \frac{\mathrm{Cov}(Z_m, Z_n)}{\sqrt{\mathrm{Var}(Z_m)\mathrm{Var}(Z_n)}} = \mu^{n-m}\sqrt{\frac{\mathrm{Var}(Z_m)}{\mathrm{Var}(Z_n)}}.$$

If $\mu \neq 1$, then

$$\mathrm{Var}(Z_n) = \sigma^2 \mu^{n-1}\frac{\mu^n - 1}{\mu - 1},$$

so

$$\mathrm{Corr}(Z_m, Z_n) = \mu^{(n-m)/2}\sqrt{\frac{\mu^m - 1}{\mu^n - 1}}.$$

If $\mu = 1$, then $\mathrm{Var}(Z_n) = n\sigma^2$, and hence

$$\mathrm{Corr}(Z_m, Z_n) = \sqrt{\frac{m}{n}}.$$

(b) Step 1: Condition on Z_m.
If $Z_m = k$, then there are exactly k individuals at generation m. Each one produces an independent subtree whose expected size after $n - m$ more generations is μ^{n-m}.

Step 2: Add the expectations of the k subtrees.
Therefore,

$$\mathbb{E}(Z_n \mid Z_m = k) = k\,\mu^{n-m}.$$

Final answers:

$$\boxed{\mathbb{E}(Z_n Z_m) = \mu^{n-m}\mathbb{E}(Z_m^2), \qquad \mathrm{Cov}(Z_m, Z_n) = \mu^{n-m}\mathrm{Var}(Z_m), \qquad \mathbb{E}(Z_n \mid Z_m = k) = k\mu^{n-m}.}$$

Question

Problem 5. Find the extinction probability ρ for a branching process with offspring distribution:

(a) (i) $\mathbb{P}(\xi = 0) = \frac{2}{3}$, $\mathbb{P}(\xi = 3) = \frac{1}{3}$.
(ii) $\mathbb{P}(\xi = 0) = \frac{1}{2}$, $\mathbb{P}(\xi = 1) = \mathbb{P}(\xi = 3) = \frac{1}{4}$.

(b) The offspring distribution is Poisson(λ). Determine the critical value λ_c below which extinction occurs with probability 1. Find the extinction probability ρ for $\lambda > \lambda_c$.

(c) Suppose

$$G(s) = p_0 + p_1 s + p_2 s^2, \qquad p_2 > 0, \quad p_k = 0 \ (k \geq 3), \quad 2p_2 + p_1 > 1.$$

Show that the extinction probability is

$$\rho = \frac{p_0}{p_2}.$$

(d) A branching process starts with $Z_0 = N$ individuals, each evolving independently with the same offspring distribution. Show that the probability of ultimate extinction is ρ^N.

Theory used

The extinction probability is the smallest fixed point of $G(s) = s$ in $[0, 1]$. Also, for Poisson(λ) offspring,

$$G(s) = e^{\lambda(s-1)}.$$

The mean offspring number decides the criticality: extinction occurs a.s. when $\mu \leq 1$, and survival has positive probability when $\mu > 1$.

Step-by-step solution

(a)(i) Step 1: Write the pgf.

$$G(s) = \frac{2}{3} + \frac{1}{3}s^3.$$

Step 2: Solve $G(s) = s$.
We need

$$\frac{2}{3} + \frac{1}{3}s^3 = s.$$

Multiply by 3:

$$2 + s^3 = 3s \quad \Longleftrightarrow \quad s^3 - 3s + 2 = 0.$$

Factor:

$$s^3 - 3s + 2 = (s-1)^2(s+2).$$

The fixed points in $[0, 1]$ are therefore $s = 1$ only.
Hence

$$\boxed{\rho = 1.}$$

(a)(ii) Step 1: Write the pgf.

$$G(s) = \frac{1}{2} + \frac{1}{4}s + \frac{1}{4}s^3.$$

Step 2: Solve $G(s) = s$.
We need

$$\frac{1}{2} + \frac{1}{4}s + \frac{1}{4}s^3 = s.$$

Multiply by 4:

$$2 + s + s^3 = 4s \quad \Longleftrightarrow \quad s^3 - 3s + 2 = 0.$$

Again,

$$s^3 - 3s + 2 = (s-1)^2(s+2).$$

So the only root in $[0, 1]$ is 1. Therefore

$$\boxed{\rho = 1.}$$

(b) Step 1: Write the pgf for Poisson offspring.
If $\xi \sim \text{Poisson}(\lambda)$, then

$$G(s) = \exp(\lambda(s-1)).$$

Step 2: Determine the critical value.
The mean offspring number is

$$\mu = G'(1) = \lambda.$$

Hence the critical value is

$$\boxed{\lambda_c = 1.}$$

Step 3: Solve the extinction equation.
The extinction probability ρ satisfies

$$\rho = e^{\lambda(\rho-1)}.$$

This equation always has the trivial solution $\rho = 1$. When $\lambda > 1$, there is a second solution in $(0, 1)$, and the extinction probability is the smaller fixed point.
If one wants an explicit formula, rewrite as

Question

Problem 6. Let the total progeny of the branching process be

$$S = \sum_{n=0}^{T} Z_n,$$

where $Z_0 = 1$ and T is the extinction time. Assume $T < \infty$ almost surely.

(a) Show that the generating function $H(s) = \mathbb{E}(s^S)$ satisfies

$$H(s) = s\,G(H(s)), \qquad 0 < s \leq 1,$$

where G is the offspring pgf. Show that this equation has a unique solution when $0 \leq s \leq 1$.

(b) Show that when

$$G(s) = q + ps^2, \qquad p \leq \frac{1}{2},$$

the total progeny has generating function

$$H(s) = \frac{1 - \sqrt{1 - 4pqs^2}}{2ps}.$$

Theory used

For the total progeny S, the key identity is:

$$S = 1 + \sum_{i=1}^{\xi} S_i,$$

where, conditionally on the first generation size ξ, the variables S_i are i.i.d. copies of S. This immediately gives the functional equation $H(s) = sG(H(s))$. When G is quadratic, the equation becomes a quadratic equation in $H(s)$.

Step-by-step solution

(a) Step 1: Decompose the total progeny.
The initial individual contributes 1 to the total progeny. If it has ξ children, then each child starts an independent copy of the same branching process. Hence

$$S = 1 + S_1 + \cdots + S_\xi,$$

where, conditionally on ξ, the random variables $S_1, S_2, \ldots$ are i.i.d. copies of S.

Step 2: Take the generating function.
Let

$$H(s) = \mathbb{E}(s^S).$$

Then

$$H(s) = \mathbb{E}\Big(s^{\,1+S_1+\cdots+S_\xi}\Big) = s\,\mathbb{E}\Big(s^{S_1+\cdots+S_\xi}\Big).$$

Conditioning on ξ, and using independence of the S_i's,

$$\mathbb{E}\Big(s^{S_1+\cdots+S_\xi} \mid \xi\Big) = H(s)^\xi.$$

So

$$H(s) = s\,\mathbb{E}(H(s)^\xi) = s\,G(H(s)).$$

Thus the functional equation is

$$\boxed{H(s) = s\,G(H(s)).}$$

Step 3: Why the solution is unique on $[0, 1]$.
For fixed $s \in [0, 1)$, the map $x \mapsto sG(x)$ sends $[0, 1]$ into itself. Also, since G is a pgf, it is increasing and convex, and for branching processes with a.s. extinction we have $\mu = G'(1) \leq 1$. Hence

$$0 \leq \frac{d}{dx}(sG(x)) = sG'(x) \leq sG'(1) \leq s < 1.$$

So $x \mapsto sG(x)$ is a contraction on $[0, 1]$, which gives a unique fixed point there.

Conclusion for (a):

$$\boxed{H(s) = sG(H(s)), \qquad 0 \leq s \leq 1, \text{ with a unique solution in } [0,1].}$$

(b) Step 1: Substitute the given pgf.
Here

$$G(s) = q + ps^2, \qquad p \leq \frac{1}{2}, \quad q = 1 - p.$$

Therefore the functional equation becomes

$$H(s) = s\Big(q + pH(s)^2\Big).$$

Step 2: Rearrange into a quadratic.
Move all terms to one side:

$$ps\,H(s)^2 - H(s) + qs = 0.$$

Step 3: Solve the quadratic equation.
Using the quadratic formula,